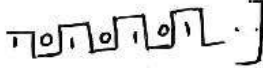


SAMPLING & QUANTIZATION

4.1 INTRODUCTION:

- ✓ Digital as well as Analog system, both are used to transmit signals from one place to another.
- ✓ In Analog communication the transmission of information is done in analog form. [continuous signal ]
 - Analog signals are continuous signals, represented by sine waves.
 - Ex: Human voice, Natural sound } $\Rightarrow$ has continuous range of values
- ✓ In Digital communication the transmission of information is done in digital form. [Discrete signal ]
 - Digital signals are, Discrete signals, represented by square wave.
 - Ex: computers, optical drives } $\Rightarrow$ has discontinuous values
- The effect of Distortion, Noise and Interference is much less in digital signals. Discrete circuits are more reliable, easy to design and cheaper than analog circuits.
- The first step in digital communication system is the conversion of analog signal to a digital/discrete time signal.

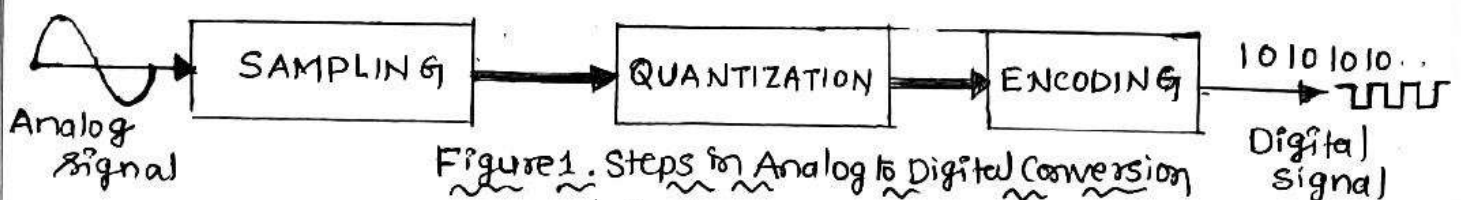


Figure 1. Steps in Analog to Digital Conversion

The key steps involved in the conversion of Analog signal to Digital signal are

1. SAMPLING
2. QUANTIZATION
3. ENCODING

* The complete details of sampling, Quantization techniques, Encoding techniques are presented in this module and continued in module-5.

4.2. Why digitize Analog Sources:-

* VTU MODEL QP *
6M.

The advantages of Using DIGITAL SIGNALS over ANALOG are:

1. The effect of Distortion, Noise and Interference is much less for digital signals.
2. Discrete circuits are more reliable, easier to design & cheaper than analog circuits.
3. With digital systems, it is easier to integrate different services.

Ex: Video + Sound track
1 2

4. Digital signals are simpler to characterize and typically do not have the same amplitude range and variability as Analog signals. [either 1 or 0]
5. Digital circuits are less sensitive to physical effects such as Vibration & Temperature.
6. Digital systems are less sensitive to NOISE than analog.
7. The transmission scheme is relatively independent of the source.
8. Multiplexing techniques, are more easily implemented with digital transmission strategies.
9. There are techniques for removing redundancy / Adding redundancy to a digital transmission. Such that the errors that occur during transmission may be corrected at the receiver.
10. High Security.
11. In Digital systems the error correction & Detection techniques can be implemented easily.
12. Techniques such as EQUALIZATION, especially Adaptive versions, are easier.

4.2.1: COMPARISON BETWEEN ANALOG & DIGITAL COMMUNICATION

PARAMETER	ANALOG SYSTEM	DIGITAL SYSTEM
1. Immunity to NOISE	• LESS	✓ MORE
2. System Complexity, Implementation	• MORE	✓ LESS
3. System Cost	• MORE	✓ LESS
4. Error Detection & Correction	• Not possible	✓ possible
5. Information Security	• less	✓ High (∴ suitable for military applications)
6. Multiplexing Technique used	• FDM	✓ TDM
7. Bandwidth Requirement - nt	• Less	Due to higher bit rates, <u>higher</u> Channel Bandwidth is required.
8. REPEATERS Usage	It is not possible to separate NOISE & SIGNAL. Therefore REPEATERS cannot be used.	✓ REPEATERS can be used.
9. Programming / coding	• Not possible	✓ possible.
10. Modulation Techniques	• AM, FM, phase-Mod	✓ ASK • FSK • PSK • PPM • PWM • PAM
11. SYNCHRONIZATION	• not required	✓ required
12. Quality of RECONSTRUCTION	• GOOD	✓ Excellent

4.3: SAMPLING PROCESS:

VTU MODEL QP*

Q) State SAMPLING THEOREM and explain the same with sketches and equations. 7 Marks.

Statement: SAMPLING THEOREM, states that "Any continuous time base band signal can be completely represented by its samples and recovered from its samples if the sampling frequency (f_s) is greater than or equal to twice the highest frequency present in the base band signal".

i.e., Sampling frequency,

$$f_s \geq 2W$$

Where, W = Maximum frequency present in base band signal.

$$W = \text{Max}(f_{m_1}, f_{m_2}, f_{m_3}, \dots)$$

proof:-

↳ consider an arbitrary signal $x(t)$ of finite energy, which is specified for all time instants.

↳ A segment of the signal $x(t)$ is shown in Figure 1(a)

↳ Let $S_g(t)$ represents the periodic impulse train and is given by: $S_g(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$, shown in Figure 1(b).

Where n , takes all possible values of integers.

$$T_s = \text{Sampling period} = \frac{1}{f_s}$$

↳ The reciprocal of Sampling period (T_s) is called sampling rate, $f_s = \frac{1}{T_s}$.

↳ The Instantaneous sampling of signal $x(t)$ is indicated in Figure 1(c).

Figure 1(a):
portion of $x(t)$

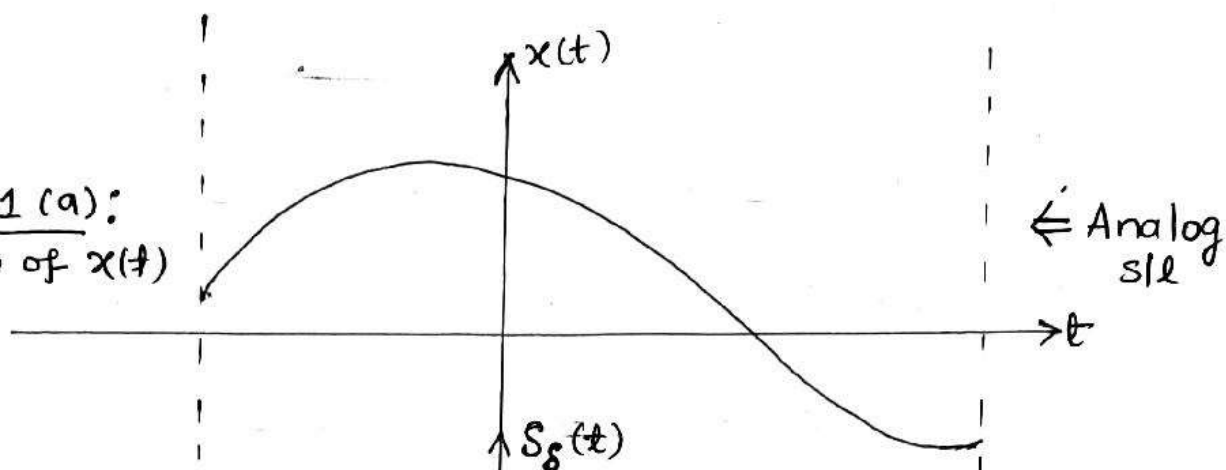


Figure 1(b):
Train of Impulses
 $S_g(t)$

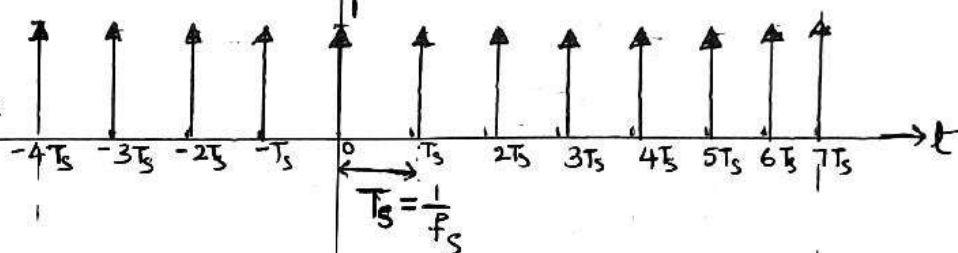
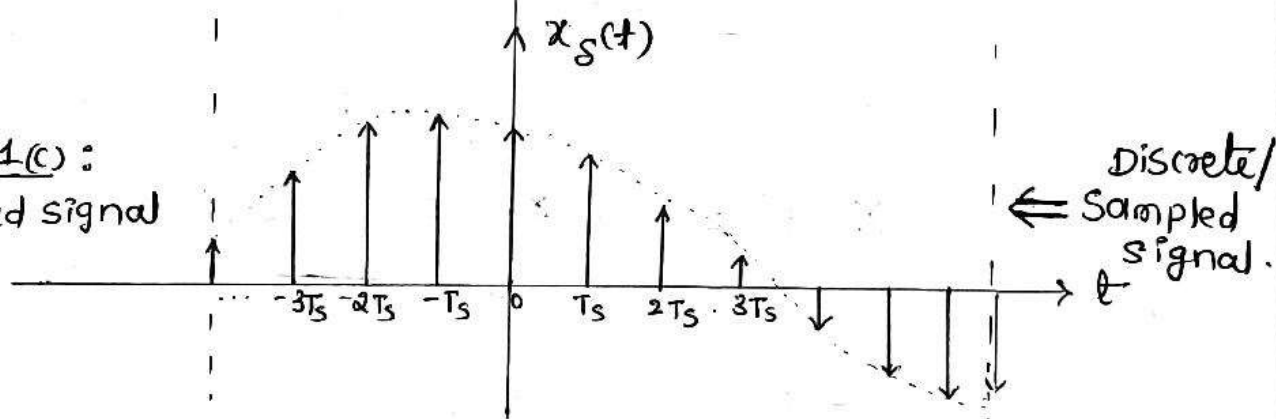


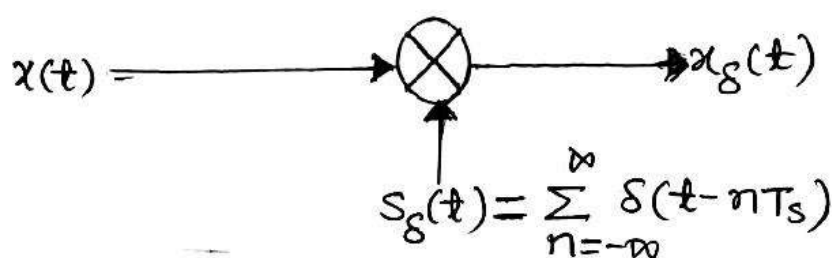
Figure 1(c):
Sampled signal
 $x_g(t)$



Let $x_g(t)$ denotes the signal obtained by multiplying $S_g(t)$ and instantaneous value of $x(t)$ at every T_s seconds.

∴ The sampled signal: $x_g(t)$ is given by,

$$x_g(t) = x(t) \cdot S_g(t) \longrightarrow (1)$$



substitute $S_g(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$ $\rightarrow (2)$ } Train of Impulses

$$\therefore x_g(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(t) \cdot \delta(t - nT_s)$$

Using shifting property of impulse function,

W.K.T. $x(t) \cdot \delta(t - nT_s) = x(nT_s) \cdot \delta(t - nT_s)$

$$\therefore \boxed{x_g(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \cdot \delta(t - nT_s)} \quad \leftarrow \begin{array}{l} \text{sampled} \\ \text{version of } x(t) \end{array} \rightarrow (3)$$

In Frequency domain,

consider, $x_g(t) = x(t) \cdot S_g(t)$ —

Taking Fourier transform on both sides we get

$$X_g(f) = X(f) * S_g(f) \quad \left[\begin{array}{l} \because \text{Multiplication in} \\ \text{time domain is equal} \\ \text{to CONVOLUTION (*)} \\ \text{in Frequency domain} \end{array} \right] \rightarrow (4)$$

Where,

$$S_g(f) = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s) \rightarrow (5)$$

Using (5) in (4),

$$X_g(f) = X(f) * f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

$$X_g(f) = f_s \left\{ X(f) * \sum_{n=-\infty}^{\infty} \delta(f - nf_s) \right\} \rightarrow (6)$$

From Convolution property of impulse function,

W.K.T., $X(f) * \delta(f - nf_s) = X(f - nf_s)$

$$\therefore (6) \Rightarrow X_g(f) = f_s \cdot \sum_{n=-\infty}^{\infty} \underbrace{X(f) * \delta(f - nf_s)}_{X(f - nf_s)}$$

$$X_g(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s) \rightarrow (7)$$

Equation (7) can be re-written as

$$X_S(f) = \underbrace{f_s X(f)}_{\eta=0} + f_s \sum_{\substack{\eta=-\infty \\ \eta \neq 0}}^{\infty} X(f - \eta f_s) \rightarrow (8)$$

When the spectrum of $X_S(f)$ is passed through an LPF then the second term in RHS of equation (8) is eliminated & results in,

$$X_S(f) = f_s \cdot X(f)$$

$$\therefore X(f) = \frac{1}{f_s} \cdot X_S(f) \rightarrow (9)$$

$-W < f < W$

Where $f_s = 2W = 2 [\underbrace{\text{Max}(f_{m1}, f_{m2}, \dots)}_{W}]$

∴ "Band limited signal can be RECOVERED from sampled version," $x_s(t)$

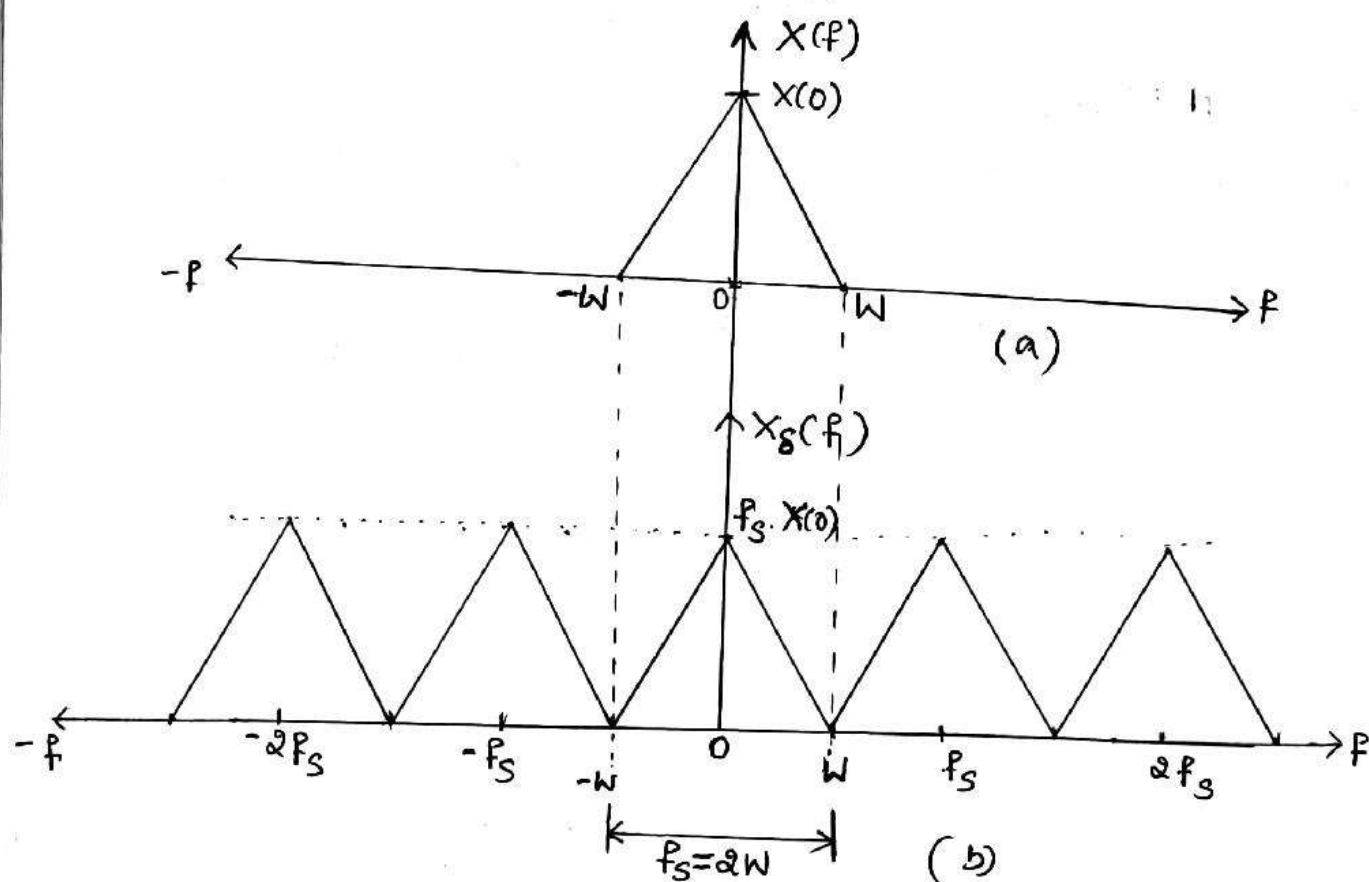


Fig (2): (a) Spectrum of strictly band limited signal $x(t)$

(b) Spectrum of a sampled version of $x(t)$ for $T_s = \frac{1}{f_s} = \frac{1}{2W}$

We may now state the Sampling theorem, in two equivalent parts:

1. A Base band signal of Finite energy, is completely described by specifying the values of the signal at instant of time separated by $T_s = \frac{1}{f_s} = \frac{1}{2W}$ seconds.
2. A band limited signal of Finite energy, may be completely recovered from a knowledge of its samples taken at the rate of " $f_s = 2W$ " samples per second.

NYQUIST RATE:-

A sampling rate of ' $2W$ ' samples per second, for a signal bandwidth of ' W ' Hertz is called Nyquist rate.

i.e., Nyquist Rate = $2W$ $\leftarrow$ It is the minimum sampling frequency required.

NYQUIST INTERVAL:-

The reciprocal of Nyquist rate is called Nyquist Interval. & is measured in seconds.

$$\text{Nyquist Interval} = \frac{1}{\text{Nyquist rate}} = \frac{1}{2W} \text{ seconds}$$

4.3.1: The Concept of UNDERSAMPLING & OVER SAMPLING

(i) When Sampling frequency $f_s = 2W$, then the type of sampling is called critical sampling / Nyquist rate sampling. Results for No-Aliasing effect & No-GUARDBAND

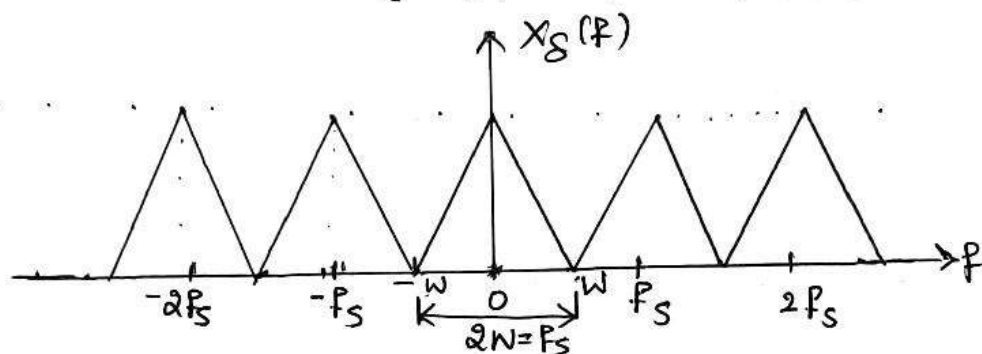


Fig 1: When $f_s = 2W$ [NYQUIST RATE]

(ii) When $f_s < 2W$:- [UNDERSAMPLING]

When $f_s < 2W$, it results in undersampling & there will be aliasing effect [overlapping]. The spectrum is shown for Fig. 2. In this case it is impossible to recover the signal.

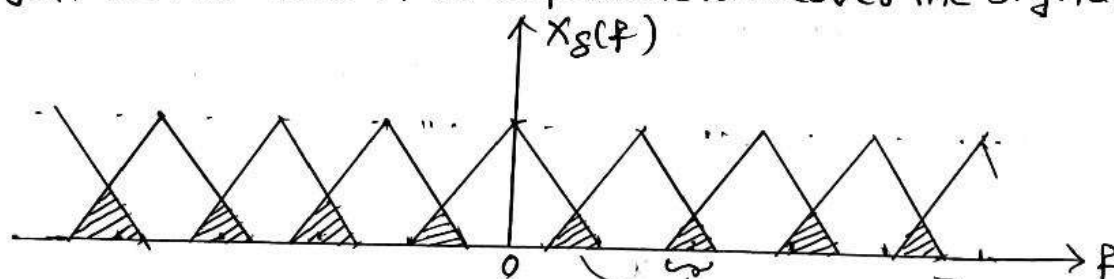
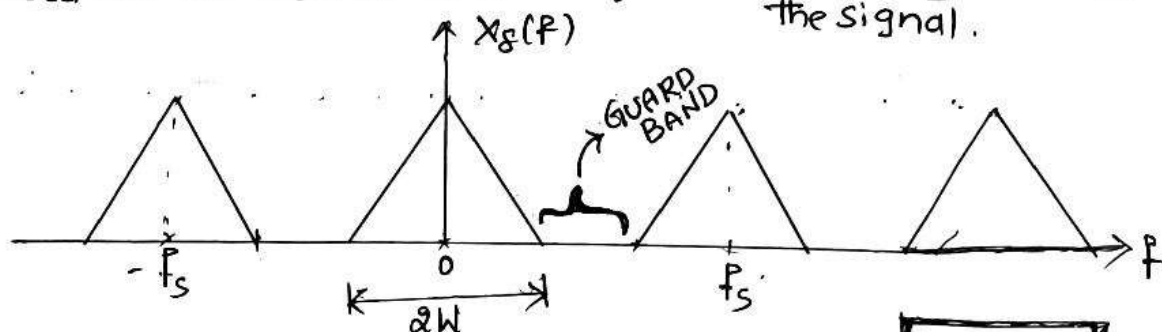


Fig 2: When $f_s < 2W$

(iii) When $f_s > 2W$:- [OVER SAMPLING]

When $f_s > 2W$, it results in over sampling & there will be Guard band b/w the adjacent spectrum of $x(t)$. It is easy to recover the signal.



∴ for Reconstruction of signal $x(t)$: Sampling frequency $f_s \geq 2W$

Numerical problems on NYQUIST RATE & NYQUIST INTERVAL:-

Formulae:

1. Nyquist rate = ' $2W$ ' samples per second @ HZ

Where $W = \text{Max}(f_{m1}, f_{m2}, \dots)$ present in given base band signal $x(t)$.

2. Nyquist Interval = $\frac{1}{\text{Nyquist rate}} = \frac{1}{2W}$, seconds.

1. Determine the Nyquist rate and Nyquist interval for: VTU 'OLD QP' June/July 2017

(i) $m(t) = \frac{1}{\pi t} [\sin(500\pi t)]$ (ii) $g(t) = \text{sinc}(200t)$

$\hookrightarrow$ (i) $m(t) = \frac{1}{\pi t} [\sin(500\pi t)]$

$\therefore \omega = 500\pi \Rightarrow 2\pi f_m = 500\pi \Rightarrow \boxed{f_m = 250 \text{ HZ} = W}$

• Nyquist rate = $2W = 2f_m = 500 \text{ HZ}$ @ samples/sec.

• Nyquist Interval = $\frac{1}{\text{Nyquist rate}} = \frac{1}{500}$ seconds

(ii) $g(t) = \text{sinc}(200t)$

W.K.T $\text{sinc}(x) = \frac{1}{\pi x} \sin(\pi x)$

$\therefore g(t) = \text{sinc}(200t) = \frac{1}{200\pi t} \sin(200\pi t)$

$\therefore \omega = 200\pi : 2\pi f_m = 200\pi \Rightarrow f_m = \frac{200}{2} = W = 100 \text{ HZ}$

$\therefore \text{Nyquist rate} = 2W = 2f_m = 2 \times 100 = 200 \text{ HZ}$

Nyquist Interval = $\frac{1}{\text{Nyquist rate}} = \frac{1}{200}$ seconds

(2) An analog signal is expressed by the equation.

$x(t) = 3 \cos(50\pi t) + 10 \sin(300\pi t) + \cos(100\pi t)$. Calculate the Nyquist rate and Nyquist Interval.

↳ Given: $x(t) = 3 \cos(50\pi t) + 10 \sin(300\pi t) + \cos(100\pi t)$

Comparing equation (1) with standard equation $\rightarrow (1)$

$$x(t) = 3 \cos(\omega_1 t) + 10 \sin(\omega_2 t) + \cos(\omega_3 t) \rightarrow (2)$$

We get

$$\omega_1 = 50\pi \quad ; \quad \omega_2 = 300\pi \quad ; \quad \omega_3 = 100\pi$$

$$2\pi f_{m1} = 50\pi \quad ; \quad 2\pi f_{m2} = 300\pi \quad ; \quad 2\pi f_{m3} = 100\pi$$

$$f_{m1} = 25 \quad ; \quad f_{m2} = 150 \quad ; \quad f_{m3} = 50$$

$$\therefore W = \max(f_{m1}, f_{m2}, f_{m3}) = \max(25, 150, 50)$$

$$W = 150 \text{ Hz}$$

$$\therefore \text{Nyquist Rate} = 2W = 2 \times 150 = 300 \text{ Hz @ samples/sec.}$$

$$\text{Nyquist Interval} = \frac{1}{\text{Nyquist rate}} = \left(\frac{1}{300} \right) = 3.333 \text{ ms}$$

(3) An analog signal is expressed by the equation

$$x(t) = \frac{1}{2\pi} \cos(4000\pi t) \cdot \cos(1000\pi t).$$

calculate the Nyquist rate and Nyquist Interval.

↳ Given: $x(t) = \frac{1}{2\pi} \cos(4000\pi t) \cdot \cos(1000\pi t)$ — (1)

W.K.T. $\cos A \cdot \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$

$\therefore (1) \Rightarrow x(t) = \frac{1}{2\pi} \left\{ \frac{1}{2} [\underbrace{\cos(3000\pi t)}_{A-B} + \underbrace{\cos(5000\pi t)}_{A+B}] \right\}$

$\therefore x(t) = \frac{1}{4\pi} \cos(3000\pi t) + \frac{1}{4\pi} \cos(5000\pi t)$ — (2)

On comparing with standard equation, we get—

$$\omega_1 = 3000\pi \quad \therefore \quad \omega_2 = 5000\pi$$

$$2\pi f_{m_1} = 3000\pi \quad \therefore \quad 2\pi f_{m_2} = 5000\pi$$

$$f_{m_1} = 1500 \quad \therefore \quad f_{m_2} = 2500$$

$\therefore W = \max(f_{m_1}, f_{m_2}) = \max(1500, 2500) = 2500 \text{ Hz}.$

$\therefore \text{Nyquist Rate} = 2W = 2 \times 2500 = 5000 \text{ Hz} \text{ @ Samples/sec.}$

$\text{Nyquist Interval} = \frac{1}{\text{Nyquist Rate}} = \frac{1}{5000} = 0.2 \text{ ms} //$

4.4 : PULSE AMPLITUDE MODULATION :- ** VTUMODEL QP⁺

(OR)

FLAT-TOP SAMPLING:

Q) Explain the generation and recovery of PAM (Flat-Top) signal with necessary equations and spectrum diagram

↳ Definition:- "It is an Analog pulse Modulation scheme in which the amplitudes of a train of rectangular carrier pulses are varied in accordance with the sample values of modulating signal."

↳ In PAM, the top of each modulated rectangular pulses is maintained flat. Hence pulse Amplitude Modulation (PAM) is also called as "FLAT TOP SAMPLING"

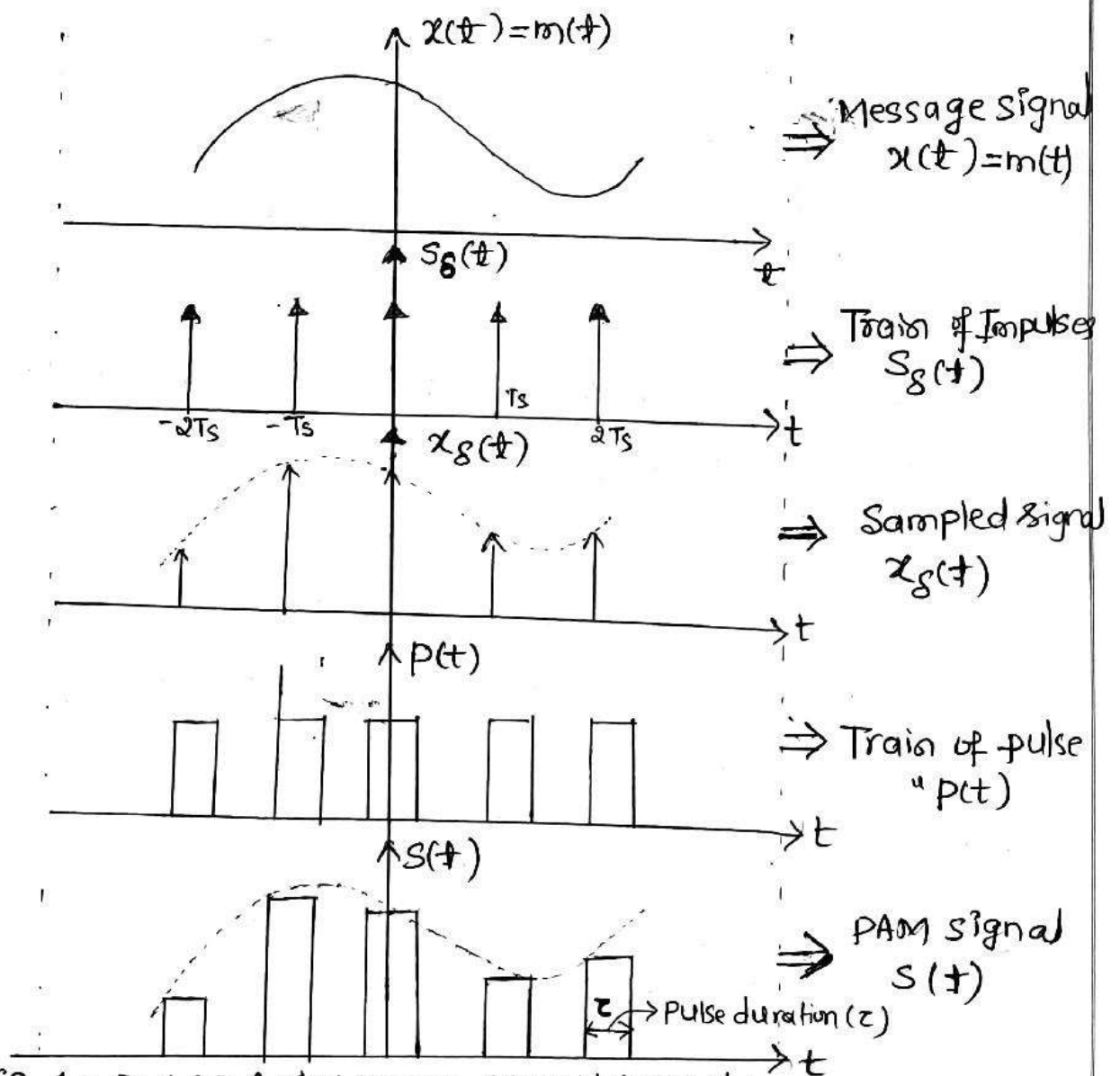


Fig. 1 : PULSE AMPLITUDE MODULATION:

- The Waveforms of a PAM signal is illustrated in Fig-1.
- ↳ Let $s(t)$ denotes the PAM signal @ Flat top sampled signal. It is expressed as

$$s(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \cdot P(t - nT_s) \quad \text{--- (1)}$$

where,

$x(nT_s)$: The sample value of $x(t)$ obtained at time $t = nT_s$.

T_s : Sampling period.

$P(t)$: Standard rectangular pulse train of duration ' τ '

Detection of PAM signal :-

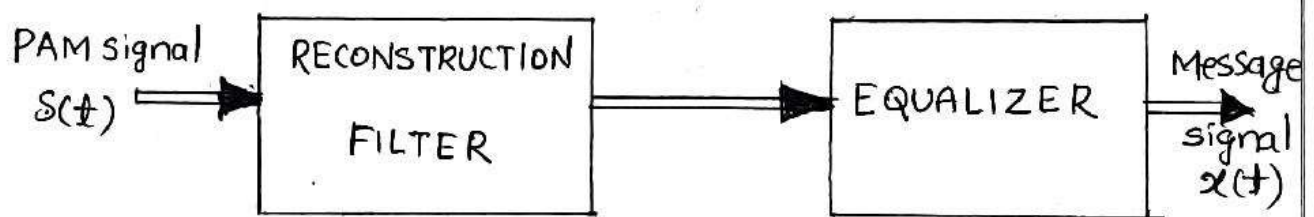


Fig. 2 : Reconstruction of message signal from PAM signal.

The Original message signal, $m(t) = x(t)$ is obtained by passing PAM signal through the reconstruction filter followed by equalizer as shown in Figure. 2.

Advantages of PAM:

- ↳ It is a base for all the digital modulation techniques.
- ↳ Transmitter & Receiver circuitry is simple and easy to construct.

Dis-Advantages of PAM:-

- ↳ It requires high BANDWIDTH for transmission.
- ↳ In PAM signal, amplitude keeps Varying, so there is a Noise associated with it.
- ↳ SNR is Less.

4.5 : TIME DIVISION MULTIPLEXING (TDM):

Q) With a neat diagram explain the concept of TDM. VTUMODEL QP* 5 Marks

→ Time Division Multiplexing is a method of transmitting and receiving independent signals over a common channel by means of synchronized switches at each end of transmission line such that each signal appears on the line only a fraction of time in an alternative pattern.

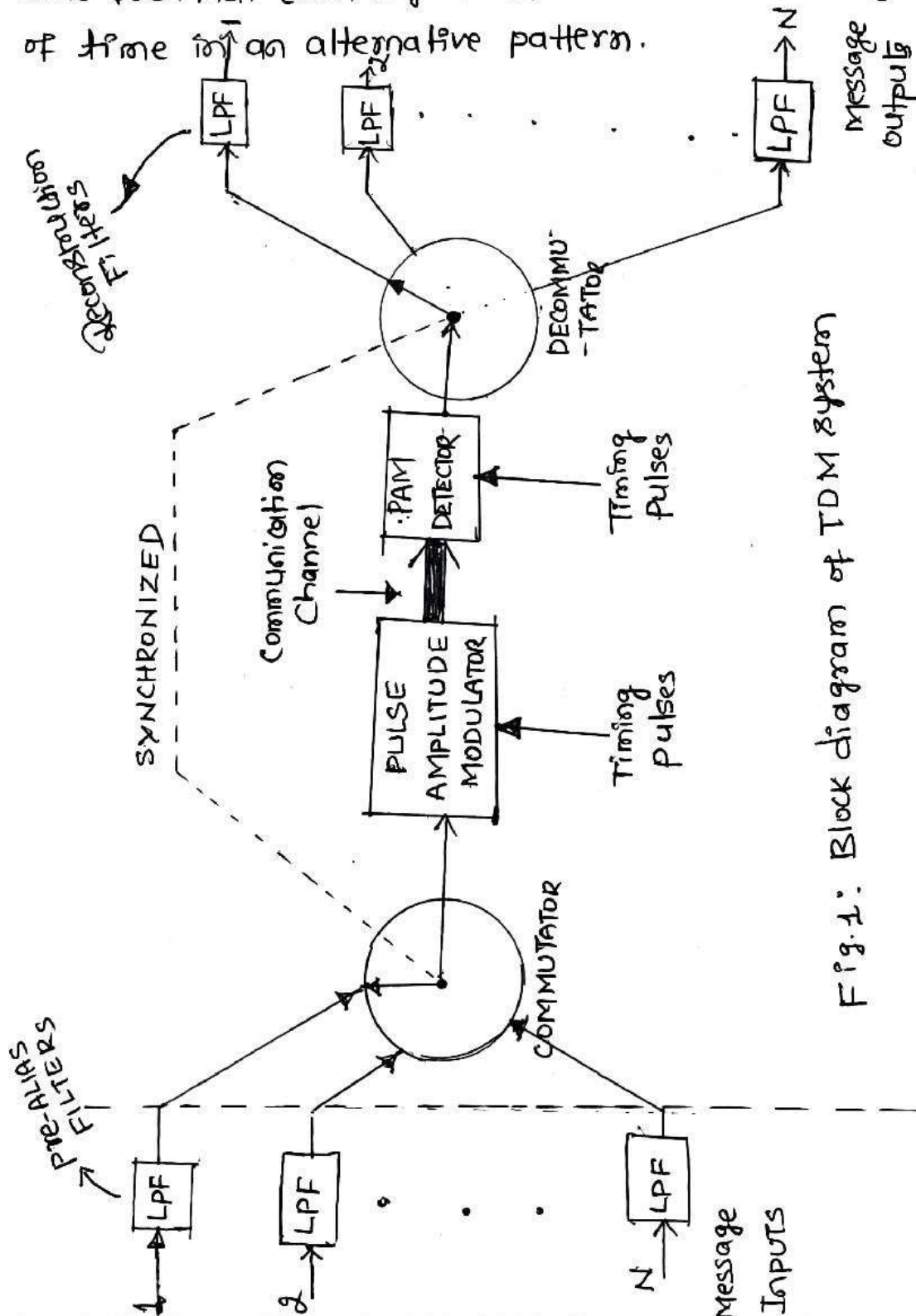


Fig. 1: Block diagram of TDM system

- Figure 1, shows the block diagram of TDM system
 - The concept of TDM is illustrated in Figure 1.
 - At Message inputs, the low pass filters are used to remove high frequency components present in the message signal.
 - The output of the PRE-ALIAS Filters are then fed to a commutator, which is usually implemented using Electronic Switching Circuitry.
- ✓ Function of COMMUTATOR:
- ↳ To collect a narrow sample in each of the N -message signals at a rate of $f_s \geq 2W$.
 - ↳ To sequentially interleave (MULTIPLEX), N -samples from each of the message signal inside a Sampling Interval $T_s = \frac{1}{f_s}$.
- The Multiplexed signal is then applied to a pulse Amplitude Modulator.
 - The pulse Amplitude Modulator, transforms the multiplexed signal into a form suitable for transmission over a Communication Channel.
 - At the Receiving end, the pulse amplitude demodulator performs the reverse operation of PAM.
 - Decommutator distributes the signals to the appropriate Low pass reconstruction filters at a rate of $f_s \geq 2W$.
 - The Decommutator operates in synchronization with the Commutator.
 - The Reconstruction filters gives the estimated message signals.

4.6: PULSE POSITION MODULATION :- *VTU MODEL QP* 5 Marks

Q) With neat block diagrams explain the Generation and Detection of PPM-Waves.

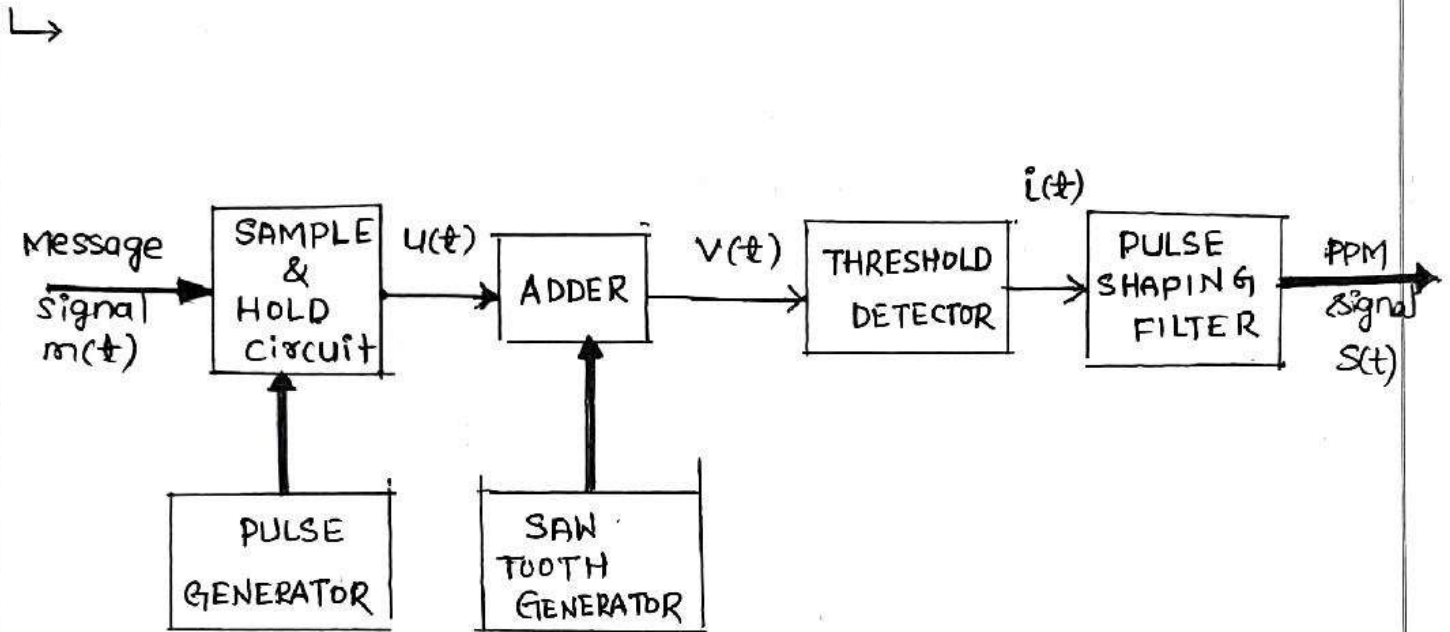


Fig.1: Generation of PPM Waves.

Generation of PPM Waves:-

The Block diagram for generating PPM Waves is shown for Fig.1.

- In pulse position Modulation (PPM), the position of a pulse relative to its unmodulated time of occurrence is varied in accordance with the message signal as shown for Fig.2(c)

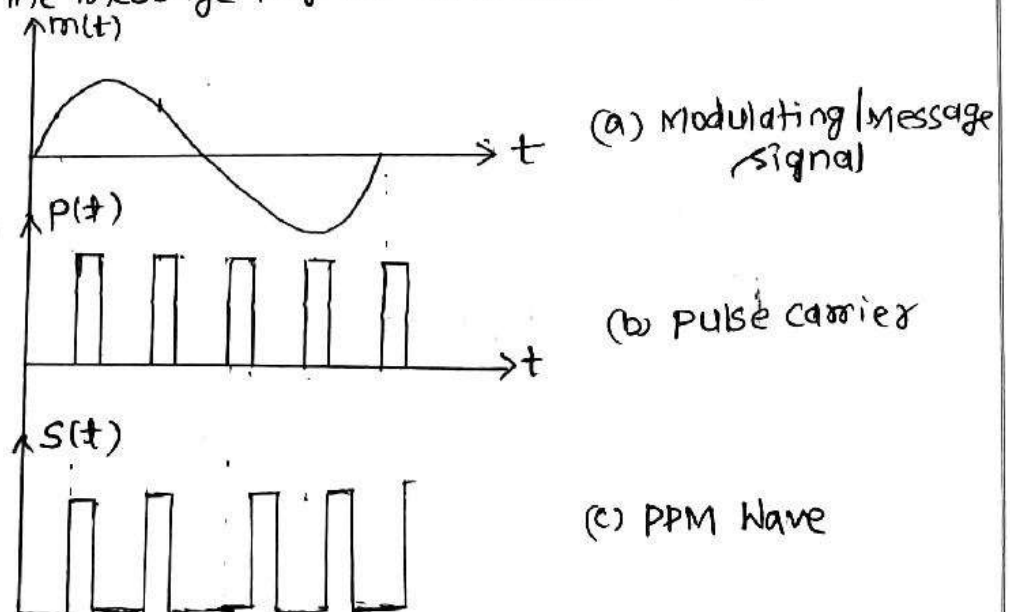


Fig.2. PPM Wave

- ↳ The Message signal $m(t)$, is first converted into a PAM signal by using SAMPLE & HOLD circuit.
- ↳ Sample & Hold circuit, gives a staircase waveform $u(t)$ as shown in figure 3(b) for the message signal $m(t)$ shown in Fig 3(a).
- ↳ The signal $u(t)$, is added to a sawtooth wave, results in the combined signal $v(t)$, as shown in Fig. 3(d).
- ↳ $v(t)$ is applied to a threshold detector that produces a very narrow pulse at each time $v(t)$ crosses zero in the Negative going direction.
- ↳ The resulting sequence of impulses, $i(t)$ is shown in Fig. 3(e).
- ↳ Finally, the PPM signal $s(t)$ is generated by using sequence of impulses, $i(t)$.
- ↳ $i(t)$ is passed through pulse shaping filter to produce PPM waves.

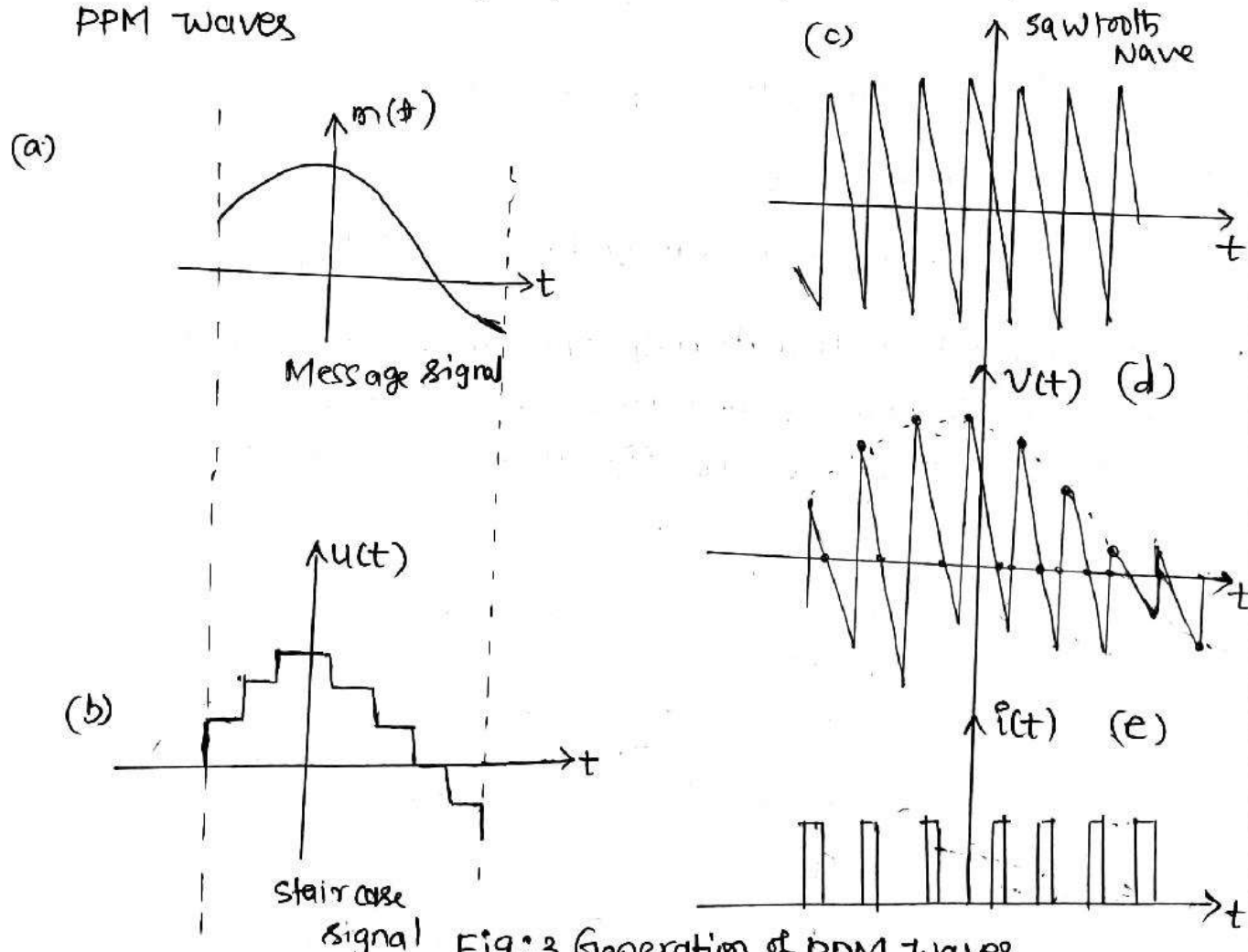


Fig: 3 Generation of PPM waves

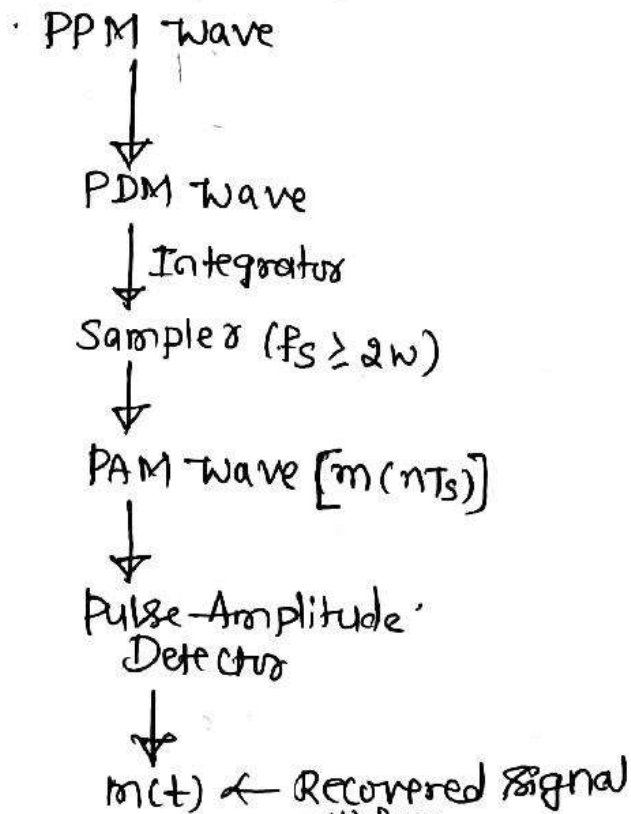
- (a) Message signal (b) Staircase signal (c) Sawtooth wave
 (d) $v(t) = (b) + (c)$ (e) PPM signal.

DETECTION OF PPM Waves:-

- Consider a ppm wave $s(t)$ with uniform sampling.
- Assume that the message signal $m(t)$ is strictly BANDLIMITED.

The function of ppm receiver is as follows:

- convert the received ppm wave into a pulse-duration modulated (PDM) wave with the same modulation.
- Integrate PDM wave using a device with a finite integration time, thereby computing the area under each pulse of the PDM wave.
- Sample the output of the integrator at a uniform rate to produce a PAM wave, whose pulse amplitudes are proportional to the signal samples $m(nT_s)$ of the original ppm wave $s(t)$.
- Demodulate the PAM wave to recover the signal $m(t)$.



- ∴ Steps in Detection of PPM Waves:-

• 4.6.1: NOISE IN PULSE POSITION MODULATION:- *VTUMODEL & P*

(Q) Describe the effect of NOISE on a pulse position Modulation system.

5 Marks

→ In a PPM system, the transmitted information is contained in the relative positions of the modulated pulses.

→ The presence of additive noise affects the performance of such a system.

- The output signal to noise ratio, assuming a full-load sinusoidal modulation is given by

$$(SNR)_0 = \frac{\pi^2 B_T T_s^2 A^2}{32 N_0} \quad \text{--- (1)}$$

- The Average noise power in a message bandwidth 'W' is equal to ' WN_0 '.

- The channel signal to noise ratio $(SNR)_c$ is:

$$(SNR)_c = \frac{3A^2}{4 T_s B_T W N_0} \quad \text{--- (2)}$$

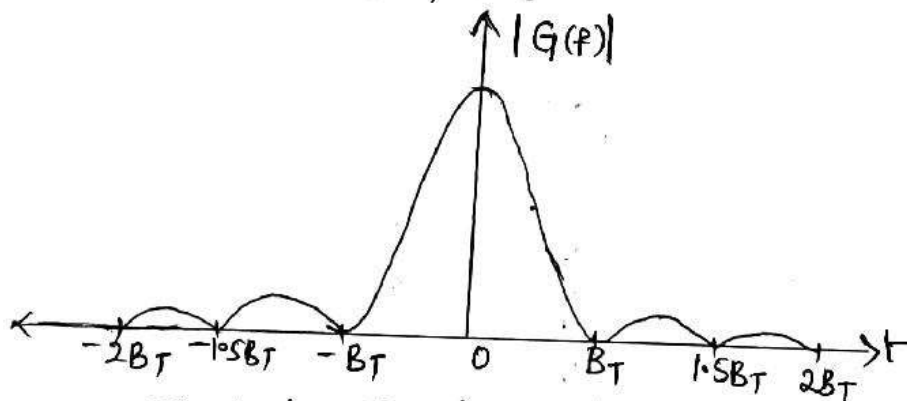


Fig :- Amplitude Spectrum of a Raised Cosine pulse

- The Figure of Merit of a ppm system using a Raised Cosine pulse as shown in Fig. is:

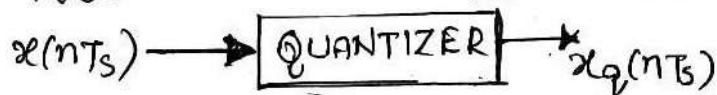
$$\text{Figure of Merit} = \frac{(SNR)_0}{(SNR)_c} = \frac{\pi^2 B_T T_s^2 A^2}{32 N_0} \times \frac{4 T_s B_T W N_0}{3 A^2}$$

$$\downarrow$$

$$(FOM)_{PPM} = \left(\frac{\pi^2}{24} \right) B_T^2 T_s^2 W$$

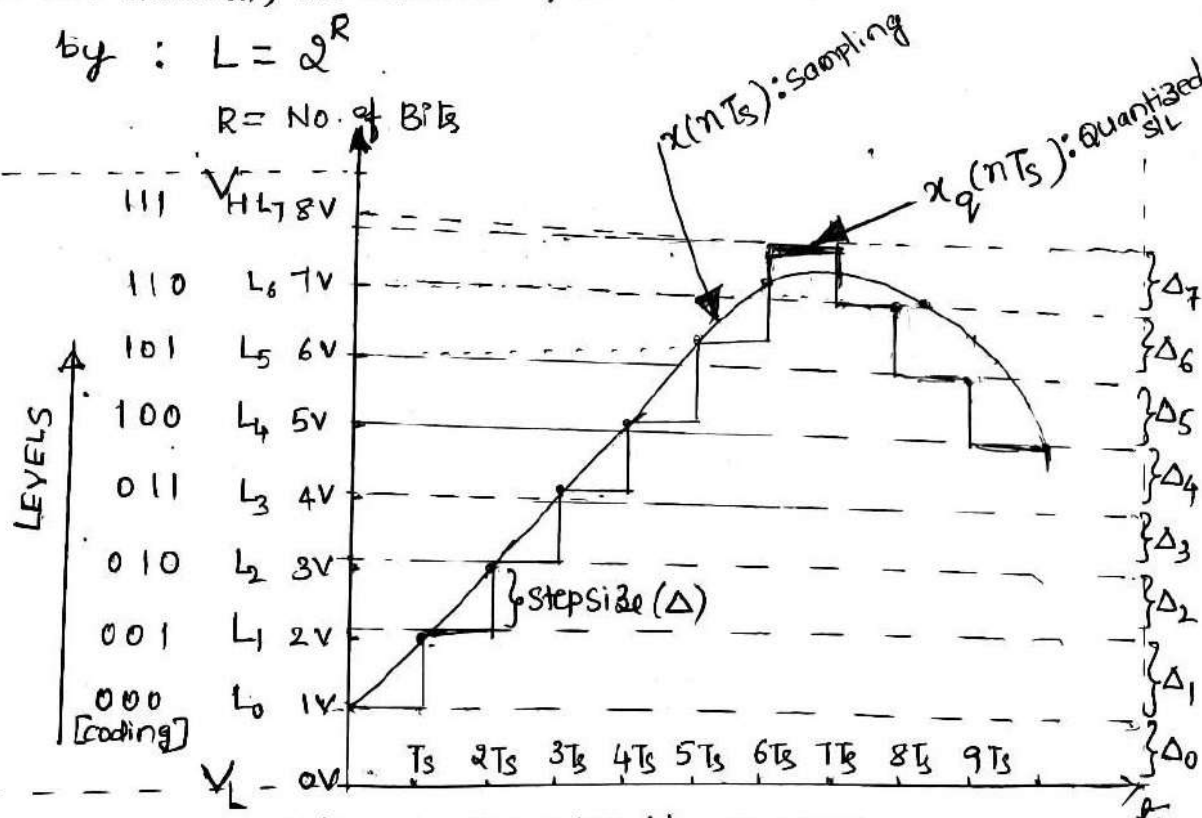
The Quantization process :-

Definition: The process of transforming sampled amplitude values of a message signal into a discrete amplitude values @ discrete amplitude levels is called as "QUANTIZATION".



Explanation:

- Consider a continuous time signal $x(t)$, whose excursions are confined to the voltage range V_L to V_H , as shown in Figure. 5.1. Let $x(nT_s)$ be the sampled version of $x(t)$.
- Let the number of required quantizer levels is '8'. [L_0 to L_7]
- In General, the Number of levels in a quantized is given by : $L = 2^R$



* Step Size is denoted by Δ and is given by

$$\Delta = \frac{V_H - V_L}{L}$$

Where $L = 2^R$ = Number of Levels, R = Number of bits in each level

$$\therefore \text{Step Size, } \Delta = \frac{V_H - V_L}{2^R}$$

* If the step size ' Δ ' is maintained same throughout the process of Quantization, then it is referred to as "UNIFORM QUANTIZATION".

* The difference between the continuous amplitude level and quantized signal level is known as "Quantization error".

i.e., Quantization error,

$$e(t) = x_q(nT_s) - x(nT_s)$$

where Quantization error varies from $+\Delta/2$ to $-\Delta/2$.

* The random errors due to quantization process produces a noise at the output of the quantizer and is referred to as "Quantization Noise".

* Consider figure 5-1, the Sampling, Quantizing and coding of an analog signal is as follows.

Sampled Values of Analog signal	2V	3V	4.2V	5.2V	6.3V	7.3V	7.6V	7V
Nearest Quantizer Level	L_1	L_2	L_3	L_4	L_5	L_6	L_7	L_7
Quantizer level Voltage	2V	3V	4V	5V	6V	7V	8V	7V
Binary code	001	010	011	100	101	110	111	110

5.1.1: Types of Quantizers:-

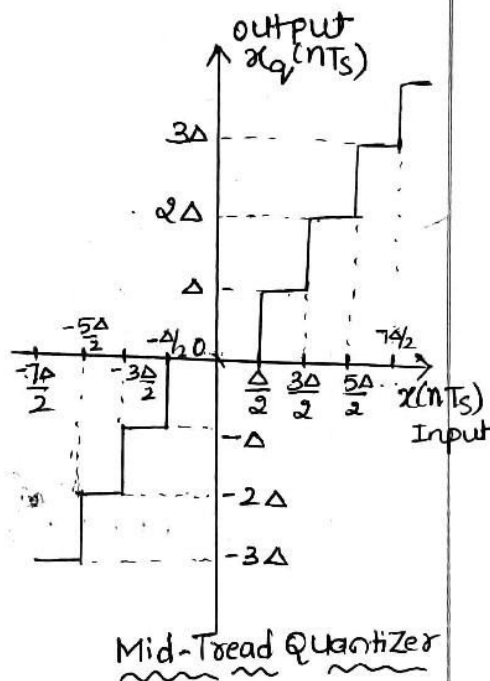
There are 2-Types of Quantizers namely,

(i) Mid-Tread type Quantizer

(ii) Mid-Raiser type Quantizer.

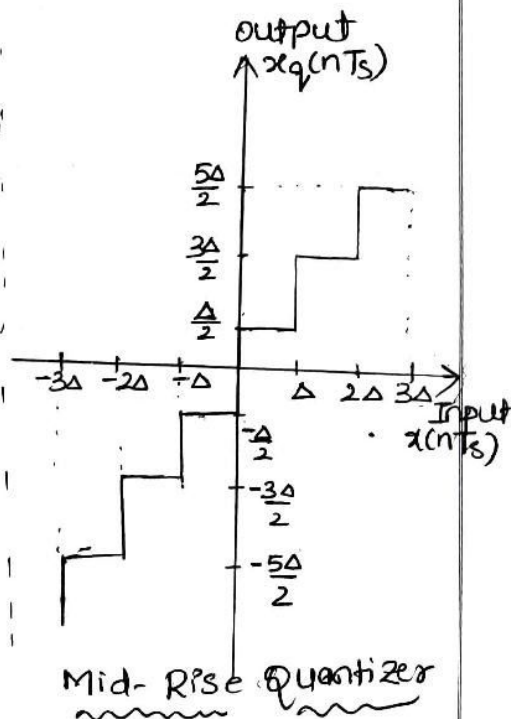
(i) Mid Tread type Quantizer:-

- In mid-tread quantizer, the decision thresholds of the quantizers are located at $\pm \frac{\Delta}{2}, \pm \frac{3\Delta}{2}, \pm \frac{5\Delta}{2}, \dots$, and
- Representation levels are located at $0, \pm \Delta, \pm 2\Delta, \pm 3\Delta, \dots$ $\Delta \Rightarrow$ step size.
- A uniform quantizer characterized in this way is referred to as a symmetric quantizer of mid-tread type.
- Origin lies in the middle of a tread of a staircase graph.
- In this type quantization levels are odd numbers.



(ii) Mid-Riser type Quantizer:-

- * In mid-riser type quantizer, the decision thresholds of the quantizers are located at $0, \pm \Delta, \pm 2\Delta, \pm 3\Delta, \dots$,
- * Representation levels are located at $\pm \frac{\Delta}{2}, \pm \frac{3\Delta}{2}, \pm \frac{5\Delta}{2}, \dots$,
- * A uniform quantizer characterized in this way is referred to as a symmetric quantizer of type mid-riser.
- * The origin lies at the middle of a riser of the staircase graph.
- * In mid-riser, quantization levels are even numbers.



5.2: Quantization Noise :- * VTUMODEL QP* 8M

(Q) Derive the expression for the output Signal to Noise Ratio of a Quantizer.



- The use of Quantization introduces an error defined as the difference between the input signal 'm' and the output signal 'v'. This error is called Quantization Noise.

$$(SNR)_0 = \frac{P}{\sigma_q^2} = \frac{\text{Signal power}}{\text{Noise power}} \rightarrow \langle A \rangle$$

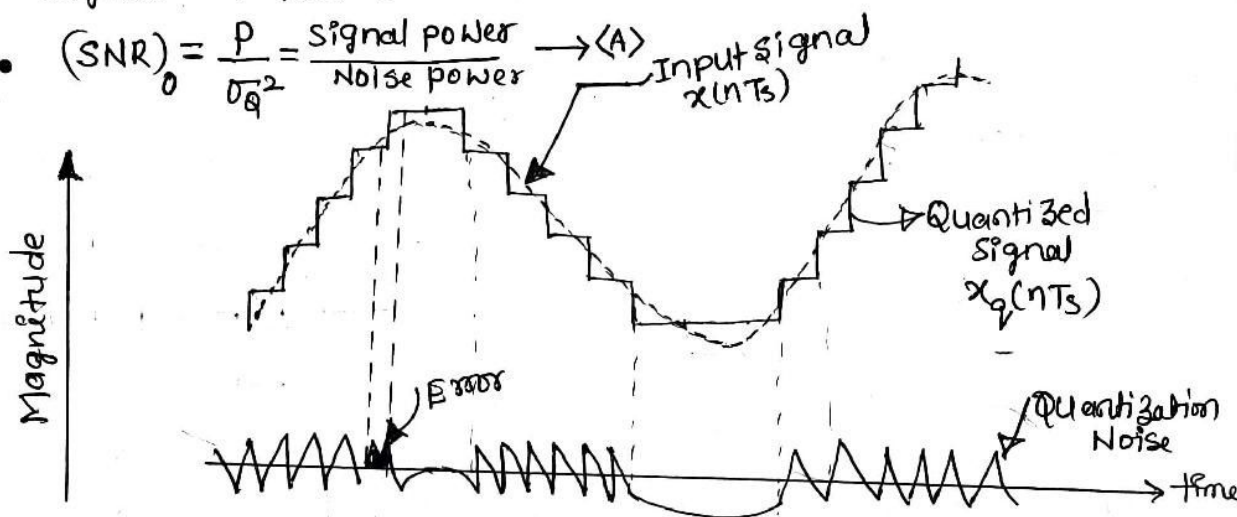


Fig. 5.2: Illustration of Quantization process & Noise.

- Fig. 5.2, shows a typical variation of the Quantization Noise as a function of time, assuming the use of a Uniform Quantizer of the mid-tread type.

- Let Random Variable 'Q' denotes the Quantization error & 'q' is its sample value.

$$\text{i.e., } q = m - v \quad \text{--- (1)}$$

where m = Input signal value, v = output signal value.

- Consider an input signal 'm' having continuous amplitude in the range $(-V_{max}, V_{max})$. Then the step size of the Quantizer is,

$$\text{Step size, } \Delta = \frac{V_H - V_L}{L} = \frac{V_{MAX} - (-V_{MAX})}{L} = \frac{2V_{MAX}}{L} \quad \text{--- (2)}$$

L = No. of Levels.

- The probability density function (pdf) of the quantization error 'Q' is as follows:

$$f_Q(q) = \begin{cases} \frac{1}{\Delta} & ; -\frac{\Delta}{2} < q \leq \frac{\Delta}{2} \\ 0 & , \text{otherwise} \end{cases}$$

- The Variance (σ_Q^2) of Quantization error is $\rightarrow (3)$

$$\sigma_Q^2 = \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} q^2 f_Q(q) dq$$

Substitute, $f_Q(q) = \frac{1}{\Delta}$

$$\therefore \sigma_Q^2 = \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} q^2 \times \frac{1}{\Delta} dq$$

$$\sigma_Q^2 = \frac{1}{\Delta} \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} q^2 dq = \frac{1}{\Delta} \left[\frac{q^3}{3} \right]_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}}$$

$$\sigma_Q^2 = \frac{1}{3\Delta} \left[q^3 \right]_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} = \frac{1}{3\Delta} \left[\frac{\Delta^3}{8} + \frac{\Delta^3}{8} \right] = \frac{1}{3\Delta} \left[\frac{\Delta^3}{4} \right]$$

$$\boxed{\sigma_Q^2 = \frac{\Delta^2}{12}} \quad \rightarrow (4)$$

σ_Q^2 : Variance, is also known as mean squared Quantization error
 (a) NORMALIZED NOISE POWER (b) Quantization noise in terms of its power.

* Let 'R' be the Number of bits per Sample. Then the Quantization Levels: $L = 2^R$ $\rightarrow (5)$

Substitute $L = 2^R$ for step size Δ , equation (2)

$$\boxed{\Delta = \frac{2V_{\max}}{2^R}} \quad \rightarrow (5)$$

substitute equation (5) in equation (4) we get

$$\sigma_q^2 = \frac{\Delta^2}{12}$$

$$\sigma_q^2 = \frac{\left(\frac{2V_{MAX}}{2^R}\right)^2}{12} = \frac{4 V_{MAX}^2}{2^{2R} \times 12} = \frac{1}{3} \cdot V_{MAX}^2 \times 2^{-2R} \quad \begin{array}{l} \text{Quantization} \\ \text{Noise} \\ \text{power} \end{array}$$

→ (6)

Let 'p' be the average power present in message signal, $m(t)$.

Then output signal-to-noise Ratio $[SNR_o]$ is:

$$(SNR)_o = \frac{\text{signal power}}{\text{Noise power}} = \frac{P}{\sigma_q^2}$$

Substitute $\sigma_q^2 = \frac{1}{3} V_{MAX}^2 \times 2^{-2R}$

$$\therefore (SNR)_o = \frac{P}{\frac{1}{3} V_{MAX}^2 \times 2^{-2R}}$$

$$(SNR)_o = \frac{3P}{V_{MAX}^2 \times 2^{-2R}}$$

$$(SNR)_o = \frac{3P}{V_{MAX}^2} \cdot 2^{2R} \quad \text{--- (7)}$$

$$\begin{aligned} 10 \log(1.5) &\approx 1.8 \\ 10 \log 2^{2R} &= 20R \cdot 10 \log 2 \\ &= 20R \log(2) \\ &= 20R(0.3010) \\ &\approx \underline{\underline{6R}} \end{aligned}$$

In terms of 'dB'

$$(SNR)_{o \text{ dB}} = 10 \log [(SNR)_o] = 10 \log \left[\frac{3P}{V_{MAX}^2} \cdot 2^{2R} \right] \quad \text{--- (8)}$$

for sinusoidal message signal: signal power $\Rightarrow P = \frac{V_{MAX}^2}{2}$

$$\therefore (8) \Rightarrow (SNR)_{o \text{ dB}} = 10 \log \left[\frac{3 \cdot (V_{MAX}^2/2)}{V_{MAX}^2} \cdot 2^{2R} \right] = 10 \log [1.5 \times 2^{2R}]$$

$$\therefore (SNR)_{o \text{ dB}} = 10 \log(1.5) + 10 \log 2^{2R} \approx (1.8 + 6R) \text{ dB}$$

5.3: PULSE CODE MODULATION :- ** VTU MODEL QP **

(Q) With a neat diagram explain the basic elements of a PCM system.

↳ In pulse code Modulation (PCM), a message signal is represented by a sequence of coded pulses, which is accomplished by representing the signal in discrete form in both time and amplitude.

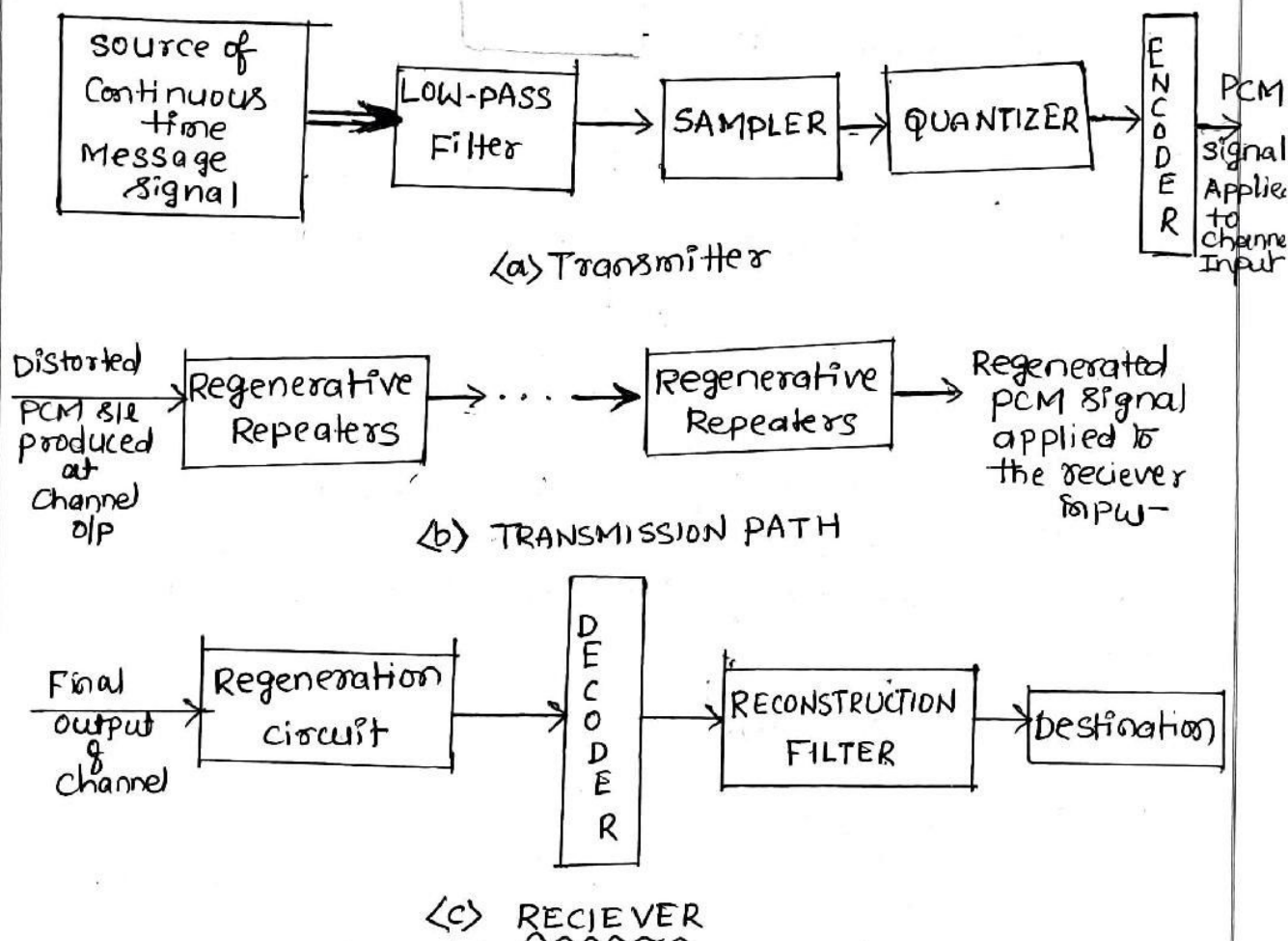


Fig. 5.3 ; The Basic elements of a PCM system.

↳ Fig. 5.3, shows the basic elements in a PCM system.

↳ The basic operations performed in the transmitter of a PCM system are Sampling, Quantization and Encoding.

↳ The block diagram of transmitter is shown in Fig. 5.3(a).

↳ The Low pass Filter prior to Sampling is included to prevent aliasing of the message signal.

↳ The Quantization and encoding operations are usually performed in the same circuit, which is called an Analog-to-DIGITAL Converter.

- The basic operations in the PCM Receiver are regeneration of Impaired signals using regeneration circuit, Decoding and reconstruction of the train of Quantized Samples.

- The block diagram of the receiver is shown in Fig. 5.3 (c).

- Regeneration also occurs at the intermediate points along the transmission path of Communication channel as shown in Fig. 5.3 (b), Transmission path.

Explanation :- (Important elements)

(i) Sampling:- According to sampling theorem
 "The sampling rate must be greater than or equal to the highest frequency component "W" of the message signal".

i.e., $f_s \geq 2W$

(ii) Quantization:-

The sampled version of the message signal is quantized. Quantizer produces a new representation of the signal that is discrete in both time and amplitude.

Types are • Mid-Tread type
 • Mid-Rise type

(iii) Encoding:-

↳ The output of Quantizer is fed to Encoder.

- The digital data can be represented by different signalling formats to transmit digital data over communication channel.
- The various signalling formats @ Waveforms @ Encoding techniques are known as Line codes
- The most commonly used line codes are :
 - <1> Unipolar NRZ
 - <2> polar NRZ
 - <3> Unipolar RZ
 - <4> Bipolar RZ
 - <5> MANCHESTER (split phase), these techniques are discussed in Next section ** 5-4

<iv>

* Regeneration:-

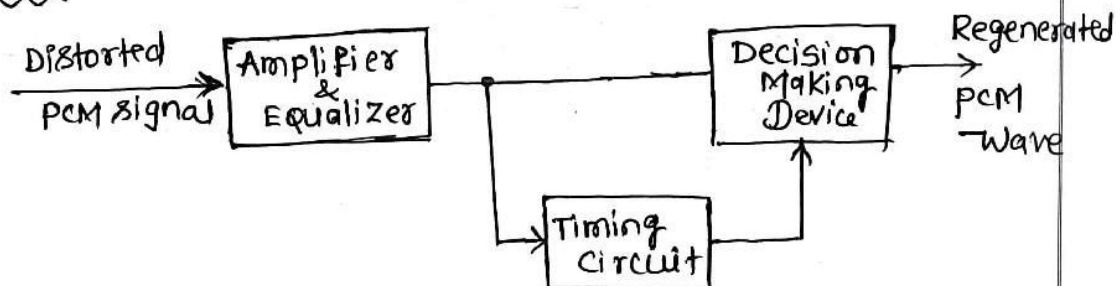


Fig. 5.4 : Block diagram of a regenerative Repeater.

* Regeneration of PCM wave is done at intermediate points along the transmission path, whenever necessary. Its block diagram is shown in Fig. 5.4.

*v> Decoding :-

• It involves the generation of a pulse. The amplitude of the decoded pulse must be equal to linear sum of all the pulses in the code word. Length of code word is $L = 2^R$
 R bits

vi> Filtering:-

• It is a process of recovering message signal. It is done by passing decoded signal through the low pass reconstruction filter.

vii> Multiplexing:- In Applications using PCM it is necessary to multiplex different message sources by TDM as discussed in Module-4.

Numerical problems on PCM & Quantization:-

List of formulae:-

(i) Sampling: $f_s = 2W$: Nyquist Rate.

(ii) codeword length: $L = 2^R$

R = Number of bits used in PCM

(iii) Quantized levels: $L = 2^R$

(iv) Quantization step size: $\Delta = \frac{V_{MAX} - V_{MIN}}{L}$

(vii) Transmission Bandwidth of PCM: $R \cdot f_s = \text{BWT}_{PCM}$ * 2

(vi) Signal to Quantization noise Ratio:-

$$(SNR)_0 = \frac{3P}{V_{MAX}^2} \times 2^{2R} \quad \text{for Sinusoidal message}$$

(vii) Bit Rate (PCM) = $R f_s$ // bps

Signal power $P = \frac{A^2}{2}$ → Amplitude.

(i) A compact Disc (CD) records audio signals digitally using PCM. Assume the audio signal bandwidth to be 15 KHz.

* VTU MODEL QP *

(a) What is the Nyquist Rate?

(b) If the Nyquist Samples are Quantized to $L = 65,536$ Levels and then binary coded. Determine the Number of bits required to Encode a sample.

(c) Assuming the signal is sinusoidal & that the Maximum signal amplitude is 1V; Determine the Quantization step & The signal to Quantization noise Ratio.

Solⁿ:- Given Audio signal bandwidth, $W = 15 \text{ KHz}$

(a) Nyquist Rate = $2W = 2 \times 15K = 30 \text{ KHz}$

(b) Given: $L = 65,536$, No. of bits $R = ?$

$$\text{W.K.T. } L = 2^R$$

Apply $\log_{10}$ on both sides

$$\log_{10}(L) = \log_{10} 2^R = R \log_{10} 2$$

$$\therefore \text{No. of bits: } R = \frac{\log(L)}{\log(2)} = \frac{\log(65,536)}{\log(2)} = 16.$$

$$\therefore R = 16 \text{ bits}$$

(c) Given: signal is sinusoidal.

$$V_{MAX} = 1V \therefore V_{MIN} = -1V.$$

Quantization step, $\Delta = ? \therefore (SNR)_0 = ?$

$$\bullet \text{ Quantization step, } \Delta = \frac{V_{MAX} - V_{MIN}}{L} = \frac{1 - (-1)}{2^R} = \frac{2}{2^8}$$
$$\Delta = 2^{-7} = 0.0078125 \text{ V/bit.}$$

• Signal to Quantization Noise Ratio:

$$(SNR)_0 = \frac{3P}{V_{MAX}^2} \times 2^{2R}$$

$$\text{Signal power: } P = \frac{V_{MAX}^2}{2} = \frac{(1)^2}{2} = 0.5 \text{ W} \therefore V_{MAX} = +1 \therefore R = 8$$

$$(SNR)_0 = \frac{3 \times 0.5}{(1)^2} \times 2^{2(8)} = 1.5 \times 2^{16}$$

$$(SNR)_0 = 98,304.$$

(2) A TV-signal with a Bandwidth of 4.2 MHz is transmitted using binary PCM. The Number of representation levels is 512. Calculate (i) code word length (ii) Final bit Rate. (iii) Transmission Bandwidth. * VTU MODEL QP *

Solution:- Given: Bandwidth of TV-signal, $W = 4.2 \text{ MHz}$

Representation levels: $L = 512$, $R = ?$ [code word length]

(i) code word length : $L = 2^R$

$$\therefore \text{code word length } R = \frac{\log(L)}{\log(2)} = \frac{\log(512)}{\log(2)}$$

$$R = 9 \text{ bits}$$

(ii) Transmission Bandwidth:- Given: $W = 4.2 \times 10^6 \text{ Hz}$

$$(BW)_{PCM} = R \times f_s = 9 \times (2W) = 9 \times 2 \times 4.2 \times 10^6$$

$$(BW)_{PCM} = 75.6 \text{ MHz}$$

(iii) Final bit rate : Bit Rate $= R f_s = 9 \times (2W)$
 $= 9 \times 2 \times 4.2 \times 10^6 = 75.6 \text{ Mbps}$

5.4: Digital Waveforms (Line codes) ** VTU MODEL QP**

<Q> What are the desirable properties of digital waveforms?

To transmit a bit sequence 10011011, draw the resulting waveforms using - Unipolar NRZ ; polar NRZ ; Unipolar RZ ; Bipolar RZ ; Manchester (split phase).

8 Marks

Definition: Digital Waveforms / Line codes:

↳ A digital waveform/line code is a pattern of voltage

& current used to represent digital data.

** properties of Digital Waveform:

↳ Digital waveform/signal is used to represent data as a sequence of discrete values.

↳ At any given instant it takes at most, one of a finite number of values. The duration of each level/bit same.

↳ It can have two possible values. Therefore it is called binary signal @ logic signal. One value is near a reference value (usually ground/0V) and the other value is near the supply voltage ($V_{CC} = 5V$).

The following FIVE Basic line codes (digital waveforms) are used to represent the digital data. They are.

<i> UNIPOLAR NRZ (Non Return to zero)

<ii> polar NRZ

<iii> Unipolar RZ (Return to zero)

<iv> Bipolar RZ

<v> Manchester (split phase)

Let us consider a bit sequence: 10011011

1. Unipolar NRZ :-

If $x(t)$ be the Digital format waveform. Then unipolar NRZ format is defined as follows:

Symbol '1': $x(t) = +A ; 0 \leq t \leq T_b$

Symbol '0': $x(t) = 0 ; 0 \leq t \leq T_b$

2. POLAR NRZ :-

Symbol '1': $x(t) = +A ; 0 \leq t \leq T_b$

Symbol '0': $x(t) = -A ; 0 \leq t \leq T_b$

3. UNIPOLAR RZ :-

Symbol '1': $x(t) = +A ; 0 \leq t \leq \frac{T_b}{2}$
 $0 ; \frac{T_b}{2} < t \leq T_b$

Symbol '0': $x(t) = 0 ; 0 \leq t \leq T_b$

4. BIPOLAR RZ :-

Symbol '1': $x(t) = +A ; 0 \leq t \leq \frac{T_b}{2}$
 $0 ; \frac{T_b}{2} < t \leq T_b$

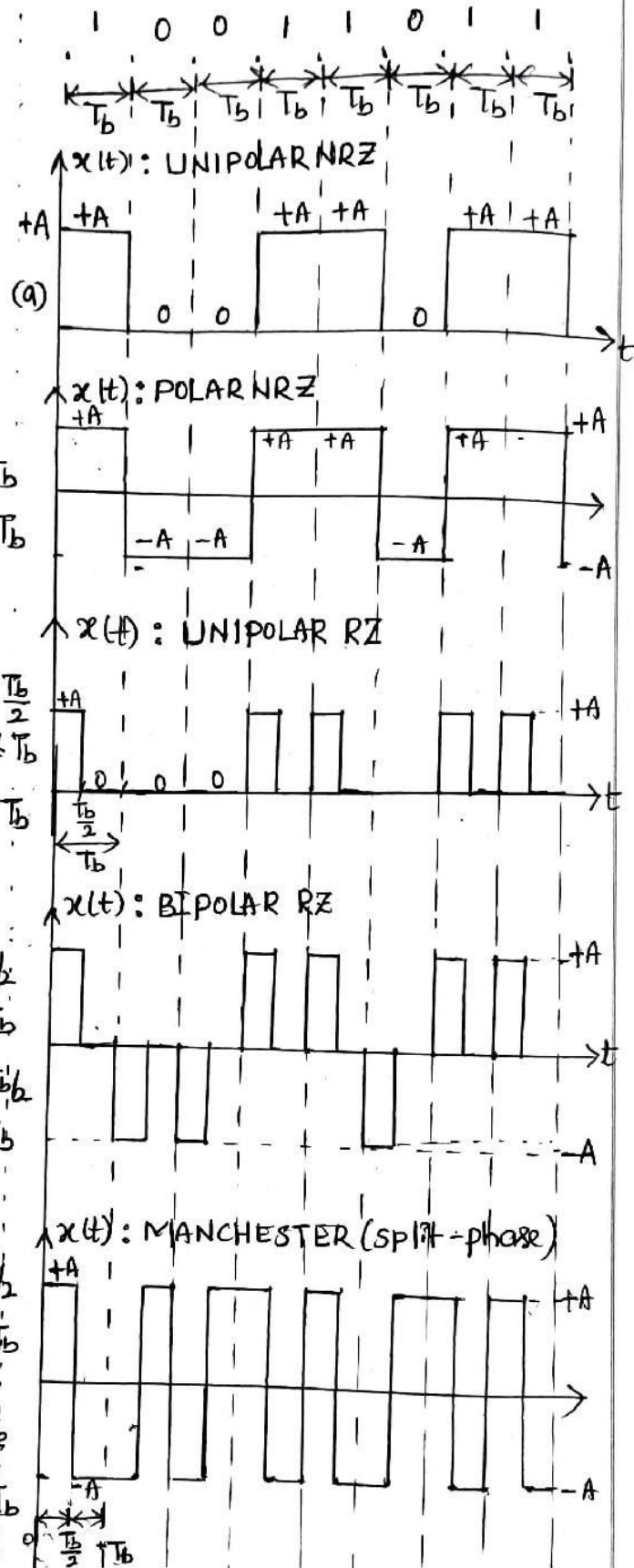
Symbol '0': $x(t) = -A ; 0 \leq t \leq \frac{T_b}{2}$
 $0 ; \frac{T_b}{2} < t \leq T_b$

5. Manchester (split phase) :-

Symbol '1': $x(t) = \begin{cases} +A & 0 \leq t \leq \frac{T_b}{2} \\ -A & \frac{T_b}{2} < t \leq T_b \end{cases}$

Symbol '0': $x(t) = \begin{cases} -A & 0 \leq t \leq \frac{T_b}{2} \\ +A & \frac{T_b}{2} < t \leq T_b \end{cases}$

Note: Using above line coding techniques, Digital w/f's can be drawn for Any given binary data.



5.5 : Delta Modulation :- (DM)

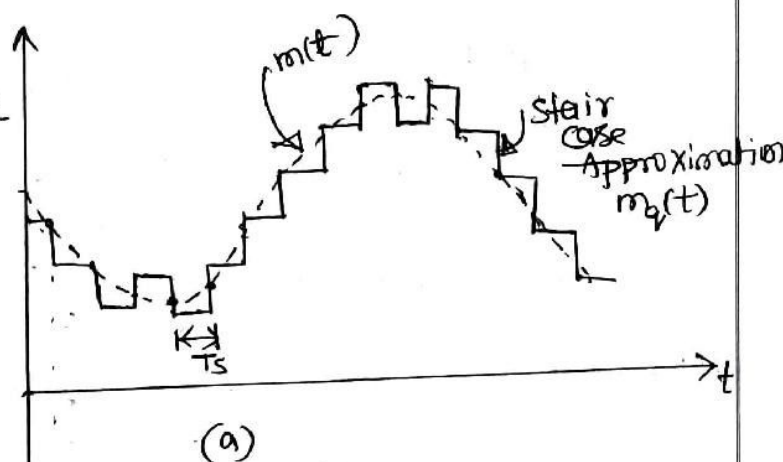
↳ In Delta Modulation, an Incoming message signal is oversampled (i.e., at a rate much higher than the Nyquist rate) to purposely increase the correlation between adjacent samples of each signal.

↳ oversampling is done to permit the use of a simple quantizing strategy for constructing the encoded signal.

↳ Delta Modulation (DM), provides a staircase approximation to the oversampled version of the message signal as shown in Fig. 5.5.

- The difference between the input and the approximation is quantized into only two levels $\pm \Delta$ corresponding to +ve & -ve differences respectively.

- It is found that staircase approximation remains within $\pm \Delta$ of the input s/l.



0 0 0 1 1 1 1 1 1 0 0 0 0
Binary Sequence at Modulator D/P

(b)

↳ Let $m(t)$ be the input s/l.

↳ $m_q(t)$ be the staircase approximation of $m(t)$.

Fig 5.5: Illustration of Delta Modulation

The basic principle of delta modulation may be formalized for the following set of discrete time relations:

$$\text{error signal : } e(nT_s) = m(nT_s) - m_q(nT_s - T_s) \quad \text{--- (1)}$$

$$\text{Quantized version of : } e_q(nT_s) = \Delta \text{sgn}[e(nT_s)] \quad \text{--- (2)}$$

$$m_q(nT_s) = m_q(nT_s - T_s) + e_q(nT_s)$$

Where T_s is the sampling period. → (3)